

Postmortem Analysis – Mobile Club Tracker

Project Name: Mobile Club Tracker

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Project Manager: Jaime Davila

Time Span: 3 months

Introduction

Mobile Club Tracker is a mobile application developed using React Native. The purpose of this application is to connect students to organizations at school and provide a place where they can explore interesting organizations or clubs, interact with them, attend their events, etc. In this application, students will be able to explore different organizations and events; follow organizations and get personalized feed; attend events by adding them to the calendar; receive push notifications. At the same time, managers will be able to post content for their organizations and interact with other users via a separate Dashboard; while administrators will be able to create new clubs and approve them in the Admin Dashboard.

What Went Well

The core functionality of the application was completed. Feed, following, club profiles, and Manager Dashboard functionalities all worked as expected. Since it was clear from the beginning what needs to be done on the backend side of the project, it turned out to be easy to implement new endpoints as needed during the course of the semester.

Dark Mode was successfully implemented using React Context, thus making it possible for users to switch between light and dark themes with one toggle. It took some time to properly organize the colors but overall it was a good decision since all colors for various elements can be found in one place.

Integration with Calendar using expo-calendar was successfully done in such a way that users can simply add an event from the Post Card or from the Detail Page of any post. Push notifications are also integrated successfully; device tokens are stored in the database, and notifications are automatically sent whenever a new post is created by the organization manager.

Another great functionality that makes the app almost complete is an Admin Dashboard where administrators can accept or reject club creation requests, along with writing optional reason for rejecting it. Everything works fine in the app without touching the database in any way.

Team members communicated pretty well and distributed their work among themselves.

Problems & What Went Wrong

The worst thing happened when we got an error while merging with the master branch in GitHub. Some changes were made to explore.jsx while the package-lock.json file had already been modified previously. As a result, we faced some issues because some of the teammates were using incorrect

imports, which referenced non-existing functions inside the auth.js file, which is why the Explore Page crashed. Luckily, it did not take too long to fix it because we simply fixed those imports, corrected some filter logic and re-generated the lock file. However, in the future, we will do some peer reviewing before committing anything to the main branch.

YourClub.jsx file crash after the third presentation. This problem occurred because there was a simple date input, where users were supposed to input the date of the event using the YYYY-MM-DD format. But somehow, we failed to check whether it worked correctly, and the date was input incorrectly, which resulted in a RangeError crash. This was fixed afterward by using DateTimePicker instead of a simple text input, which prevented entering the date wrongly.

Finally, the passwords are saved in the database without being hashed. We knew very well that passwords should be hashed and saved using bcrypt library. However, we did not have enough time for doing this because of the previous tasks.

Things To Be Changed

We will definitely add Club Tags functionality that will allow users to choose certain categories for each club and then find the desired clubs using filters. The reason why we need it is because right now, users can only find clubs using their names or descriptions, and there is nothing else they can use to look for something. If this feature were added, it could help students find clubs that interest them better. Also, tags would be useful for new students who know nothing about existing organizations at all.

We are going to set up a rule that no teammate merges his code directly to the main branch, he has to ask someone to review the changes.

We are also going to test thoroughly every scenario we are going to demonstrate during presentations.

Conclusion

Overall, the project went pretty well and we managed to achieve our initial purpose. Moreover, we also added many functionalities to our application that we did not even consider initially, including calendar integration, dark mode, notifications, per-user coloring of posts and Admin Dashboard. Of course, there were some problems, but luckily, we managed to resolve them successfully.